

Introduction To Programming Using C++ Level 2

Save Vector to File

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Definition

 `SaveVectorToFile()`

 `SaveVectorToFile()` is a function that writes the elements of a vector into a text file.

 Each vector element can be written as a separate line in the file.

Explanation

Saving Vector Data

 In the previous lesson, we loaded data from a file into a vector.

 Now we do the opposite: we take the data from the vector and write it back to a file.

Creating the Vector

 We first create a `vector<string>` containing the data we want to save.

```
vector<string> vFileContent{"Ali", "Shadi", "Maher", "Fadi", "Lama"};
```

Opening the File

 We open the file using `ios::out`.

 When an existing file is opened with `ios::out` alone, its previous contents are normally truncated before writing.

```
MyFile.open(FileName, ios::out);
```

Writing Vector Elements

 We use a range-based `for` loop to go through every element in the vector.

 `MyFile << Line << endl;` writes each element to the file on a separate line.

 The `const string&` form avoids copying each string because we only need to read it.

Closing the File

 After finishing the writing operation, we close the file.

```
MyFile.close();
```

Procedures

1. Create a `vector<string>` containing the data.
2. Pass the file name and vector to `SaveVectorToFile()`.
3. Open the file using `ios::out`.
4. Check whether the file opened successfully.
5. Loop through the vector elements.
6. Write each non-empty element to the file.
7. Close the file.

💡 Example

```
void SaveVectorToFile(string FileName, vector<string>& vFileContent)
{
    fstream MyFile;

    MyFile.open(FileName, ios::out);

    if (MyFile.is_open())
    {
        for (const string& Line : vFileContent)
        {
            if (Line != "")
                MyFile << Line << endl;
        }

        MyFile.close();
    }
}
```

💡 Example

```
int main()
{
    vector<string> vNames{"Ali", "Shadi", "Maher"};

    SaveVectorToFile("Names.txt", vNames);

    return 0;
}
```

➡ Result

When `Names.txt` is opened, it contains:

Ali

Shadi

Maher

Note

- ◆ ``ios::out`` is used when we want to write the vector data to the file.
- ◆ Opening an existing file with ``ios::out`` alone normally replaces its previous contents.
- ◆ If you want to preserve the existing contents and add new data at the end, use ``ios::app``.
- ◆ Passing the vector by reference avoids copying the entire vector. Since the function does not modify the vector, ``const`` is also appropriate.

⊘ Warning

- ◆ Writing to a file does not display the saved data on the console.
- ◆ To verify the result, open the file from its location or read it again using a file-reading function.
- ◆ Be careful when using ``ios::out`` because existing file contents may be truncated.
- ◆ If you use ``ios::app``, new data will be added instead of replacing the existing contents.



Comparison

`ios::out`

- ✓ Opens the file for writing.
- ✓ Existing contents may be truncated.
- ✓ Useful when saving the current vector as the new file contents.

`ios::app`

- ✓ Opens the file for appending.
- ✓ Preserves existing contents.
- ✓ New data is added at the end of the file.

✦ Summary

- ✓ `SaveVectorToFile()` writes vector data to a text file.
- ✓ `ios::out` is used for writing.
- ✓ A `for` loop processes each vector element.
- ✓ `<<` writes each element to the file.
- ✓ Each element can be written on a separate line.
- ✓ `close()` closes the file after writing.
- ✓ `ios::out` can replace existing file contents.
- ✓ `ios::app` can be used when we want to add data without removing existing contents.

Thank you 😊